ARTICLE · HOST–PATHOGEN PROTEIN INTERACTIONS

Coevolution Without a Common Ancestor

Reading the host–pathogen arms race from strain variation alone — which co-evolutionary signals survive when host and pathogen share no evolutionary history, and what they are worth for predicting, modeling, and folding cross-species protein interactions.

Team co-evo — liyiyuian (Ian Lee). Signal discovery → model enhancement → structure & design. Built entirely in Claude Science. Corpus of 101,132 interaction pairs · 13-species balanced benchmark · two AlphaFold-3 pilots · 20 references.

7/8
host–pathogen complexes share **zero** shared taxa — the paired alignment DCA needs can't be built

4.2×
more strain variation at SARS-CoV-2 interface residues than the rest of the domain ($p = 1 \times 10^{-4}$)

3/8
systems show the signal — it's real but pathogen-specific, not universal

0.49 → 0.80
cross-species transfer AUC climbs with phylogenetic proximity to the training set

CONTENTS

ABSTRACT

You and the virus trying to infect you share no family tree — not a distant one, none at all. Yet the standard way to read how two proteins evolve together assumes exactly the opposite: a shared ancestry deep enough to line up their sequences, residue against residue, and watch them change in concert. We tested that assumption on eight experimentally solved host–pathogen complexes. It failed on seven: the paired alignment the method needs cannot even be built, because host and pathogen share zero taxa. The method was not broken — its founding assumption was never true here. So we went looking for a co-evolutionary signal that survives without a shared tree, and found one hiding in the data that pandemic surveillance already collects: strain sequences and a single structure.

Working entirely within Claude Science, we assembled a provenance-tracked corpus of 101,132 human–pathogen interaction pairs and a structural benchmark of 8 experimentally solved complexes, and tested five hypotheses about which co-evolutionary signals survive the absence of a shared species tree. We find that classical inter-protein coupling analysis is not merely weak but undefined for 7 of 8 structural pairs, which share zero taxa; that pathogens instead concentrate strain-level variation at the host interface (SARS-CoV-2 receptor-binding residues vary 4.2× more than the rest of the domain, permutation $p = 1\times10^{-4}$), a signal computable with no paired alignment; that this signal is real but pathogen-specific (significant in 3 of 8 systems); that a lightweight arms-race feature improves cross-species interaction transfer where protein-language-model embeddings fail; and that AlphaFold-3 interface confidence does not, in a controlled pilot, discriminate interactors for a phylogenetically isolated pathogen. We position each result against the published literature, distinguishing what is established biology from what the present work adds.

BACKGROUND

1 The tool that can't be used

THE PARADOX

Reading how two proteins co-evolve normally **requires** that they share a species tree — a common ancestry deep enough to align them and watch them change together. A human receptor and the pathogen protein that grips it share **none**. The entire method rests on an assumption that host–pathogen biology violates by definition.

Co-evolution is one of the most productive ideas in structural biology. When two residues are in physical contact, a substitution at one is frequently compensated by a substitution at the other, so that across a deep alignment of homologous sequences the two positions co-vary. Direct-coupling analysis (DCA) disentangles these direct couplings from indirect correlations and recovers native residue contacts across many protein families (Morcos 2011), enabling three-dimensional structure to be computed from sequence variation alone (Marks 2011). Given a *paired* alignment (one row per species, with the two partner proteins concatenated), the same principle yields the contacts and structures of protein *complexes* (Hopf 2014), a capability since industrialized in the EVcouplings framework (Hopf 2018) and scaled to thousands of interactions inferred from co-occurring genomes (Green 2021).

The modern structure-prediction models can be read as learned co-evolution engines: AlphaFold2 (Jumper 2021), RoseTTAFold (Baek 2021), AlphaFold-Multimer (Evans 2021), and AlphaFold3 (Abramson 2024) all consume a multiple-sequence alignment and internalize the covariation signal, and ESMFold shows that a protein language model can approximate it from single sequences (Lin 2023). Downstream, AlphaFold2 with interface-confidence scoring predicts protein interactions well *within* a proteome (Bryant 2022; Humphreys 2021), and protein language models supply the mutation-effect and embedding features on which interaction classifiers are built (Meier 2021).

Every one of these methods inherits the same requirement as classical DCA: a meaningful alignment in which host and pathogen rows correspond by shared ancestry. The problem of pairing sequences *without* a species tree is therefore an active frontier, addressed by masked-language-model approaches such as DiffPALM (Lupo 2024) and the MSA Transformer (Rao 2021). For host–pathogen systems specifically, the classical route has been comparative modeling and interolog transfer (Davis 2007), supported by curated interaction resources (Navratil 2008).

In parallel, the biology of one host–pathogen interface has been characterized in exceptional detail. Deep mutational scanning of the SARS-CoV-2 receptor-binding domain mapped the constraints on folding and ACE2 binding (Starr 2020) and the complete set of mutations that escape antibody recognition (Greaney 2021), and the population-scale consequences of that variation are reviewed as spike mutation and immune escape (Harvey 2021). This establishes, independently of any prediction method, that pathogen interface residues sit under strong and variable selection.

The gap. None of the co-evolutionary machinery above runs when host and pathogen share no evolutionary history — which is precisely the host–pathogen case. A human receptor and a viral or bacterial effector cannot be placed in the same species-indexed alignment, because they have no species in common. The question this project addresses is therefore not "how well does DCA work here" but "which co-evolutionary signals exist at all when the shared-tree assumption is removed, and what are they worth for prediction, model enhancement, and structure."

HYPOTHESES

2 Five bets on what survives

The absence of a shared species tree makes some signals impossible and leaves others untouched. We formalized that intuition into five testable hypotheses, ordered so that each motivates the next.

> **H1 — the barrier.** Classical inter-protein DCA is inapplicable to host–pathogen pairs, because host and pathogen homolog sets share no species and the paired alignment the method requires cannot be constructed. The failure should be structural, not a matter of tuning, and should coexist with a working extractor on a co-speciating control.

> **H2 — a signal that survives.** A pathogen under selection to bind (or to escape immunity at) a host receptor concentrates its across-strain variation at the interface residues. Because this variation is visible across strains of a single pathogen species, it needs no host tree and no paired alignment, and should be detectable as elevated per-residue entropy at contact positions.

H3 — generality. If H2 reflects a general property of host–pathogen interfaces rather than a SARS-CoV-2 idiosyncrasy, interface-localized variation should recur across unrelated pathogen systems and across both viral and bacterial kingdoms.

H4 — model enhancement. Features derived from these co-speciation-free signals should improve cross-species interaction prediction — specifically leave-one-species-out transfer — precisely where mean-pooled protein-language-model embeddings, which encode single proteins rather than compatibility, fail.

H5 — structure. Structure-prediction interface confidence (AlphaFold-3 ipTM) should discriminate true from decoy host–pathogen pairs, and in particular might succeed for phylogenetically isolated pathogens where sequence-embedding transfer is at chance.

Two subsidiary predictions sharpen H2 and serve as controls: cross-side host ↔ pathogen coupling should be weak when the human receptor is invariant (one-sided selection), and within-pathogen strain covariation should *not* recover three-dimensional contacts, because it reflects shared immune-escape pressure rather than physical proximity — distinguishing our signal from the covariation-implies-contact premise of DCA.

METHODS

3 How we tested it

3.1 Interaction corpus

We deduplicated 130,373 evidence records from IntAct/IMEx and the structural record into 101,132 unique human–pathogen interaction pairs (87,037 human–virus, 14,095 human–bacteria), spanning 474 pathogen species, 13,676 human proteins, and 5,125 publications. Each pair carries an ordered evidence tier from co-crystallization down to yeast two-hybrid; 9,769 pairs have structural (Tier-1) evidence. This corpus defines the scope of every downstream analysis.

3.2 Structural benchmark and the co-speciation test (H1)

We selected 8 experimentally solved host–pathogen complexes spanning viral entry receptors, bacterial effectors, and toxin–receptor pairs. For each partner we built the deepest available homolog alignment and counted shared taxa between host and pathogen sides. As a positive control for the DCA extractor itself, we ran mean-field DCA on the Ras/Rho GTPase superfamily, a system that genuinely cospeciates, and scored recovered couplings against true three-dimensional contacts. Effective alignment depth (Meff) was recorded per partner to separate the depth barrier from the pairing barrier.

3.3 Interface strain variation (H2)

For SARS-CoV-2 we computed per-residue Shannon entropy across a panel of Spike strains and mapped ACE2-contact residues from the co-complex structure. The interface entropy was compared against two explicit baselines (the rest of the receptor-binding domain and the whole Spike protein), and significance was assessed with a 10,000-shuffle permutation test that randomizes the interface label across residues. Two controls accompanied the test: cross-side host↔pathogen coupling, and the Spearman correlation between within-pathogen strain covariation and three-dimensional contact.

3.4 Cross-system generality (H3)

We applied the identical entropy test to 8 systems across both kingdoms, assembling deep bacterial isolate panels (86–800 sequences) so the bacterial arm was not data-starved. Because a high fold-change on a near-invariant protein is spurious, every system was scored on both fold-change and absolute interface entropy, and significance required clearing an absolute-entropy floor in addition to the permutation test.

3.5 Cross-species prediction benchmark (H4)

Interaction pairs were represented with ESM-2 (35M) embeddings (1440-dimensional) plus sequence features, and, critically, the benchmark was rebuilt to be species-balanced: 9,172 exactly 1:1 positive/negative pairs across 13 species, capped at ≤400 positives per species so no single pathogen fold dominates the pooled metric. The classifier was a HistGradientBoostingClassifier (max_iter = 200) on standardized features. Within-species performance used stratified 5-fold cross-validation; cross-species transfer used leave-one-*species*-out (LeaveOneGroupOut), which forbids the classifier from memorizing which pathogen proteins are interaction hubs. The co-evolution contribution was an 8-dimensional block derived from the model's masked-language constraint and the host–pathogen asymmetry within it. The modeling protocol is identical across all feature sets and both cross-validation schemes, so every number is directly comparable, and pooled AUC was recomputed directly from the raw out-of-fold predictions as an independent check.

3.6 Structure-prediction pilots (H5)

Two AlphaFold-3 pilots were run on the PSC Bridges-2 cluster (account med220003p) using the project's AF3 container in a two-stage flow (MSA on CPU nodes → inference on GPU). The first was an 11-pair end-to-end validation against PDB ground truth. The second was a controlled discrimination test: for a phylogenetically isolated species (*Bacillus anthracis*, one of the four no-relative species; at LOSO AUC 0.509 it is in fact the mildest of the four, not the hardest) and a SARS-CoV-2 positive control, true and decoy host–pathogen pairs were folded and scored by interface confidence (ipTM). A compute constraint is material to interpreting the second pilot: full-database MSA search proved intractable within the allocation, so MSAs were built against UniRef90 only, which is known to depress AF3 confidence and is stated as a caveat wherever the result appears.

> *The method was not broken. Its founding assumption was never true here.*
>
> — on why classical coupling analysis fails across the host–pathogen divide

RESULTS

4 The wall, and the signal that climbs it

4.1 The pairing barrier is structural, not a matter of tuning (H1)

When we counted shared taxa across all 8 structural pairs, **7 of 8 shared exactly zero**: host and pathogen live in different kingdoms, so the pairing step that inter-protein DCA depends on has no input. This is not a weak signal but an undefined one. The single SARS-CoV-2 count of one is a database artifact, not a real shared species.

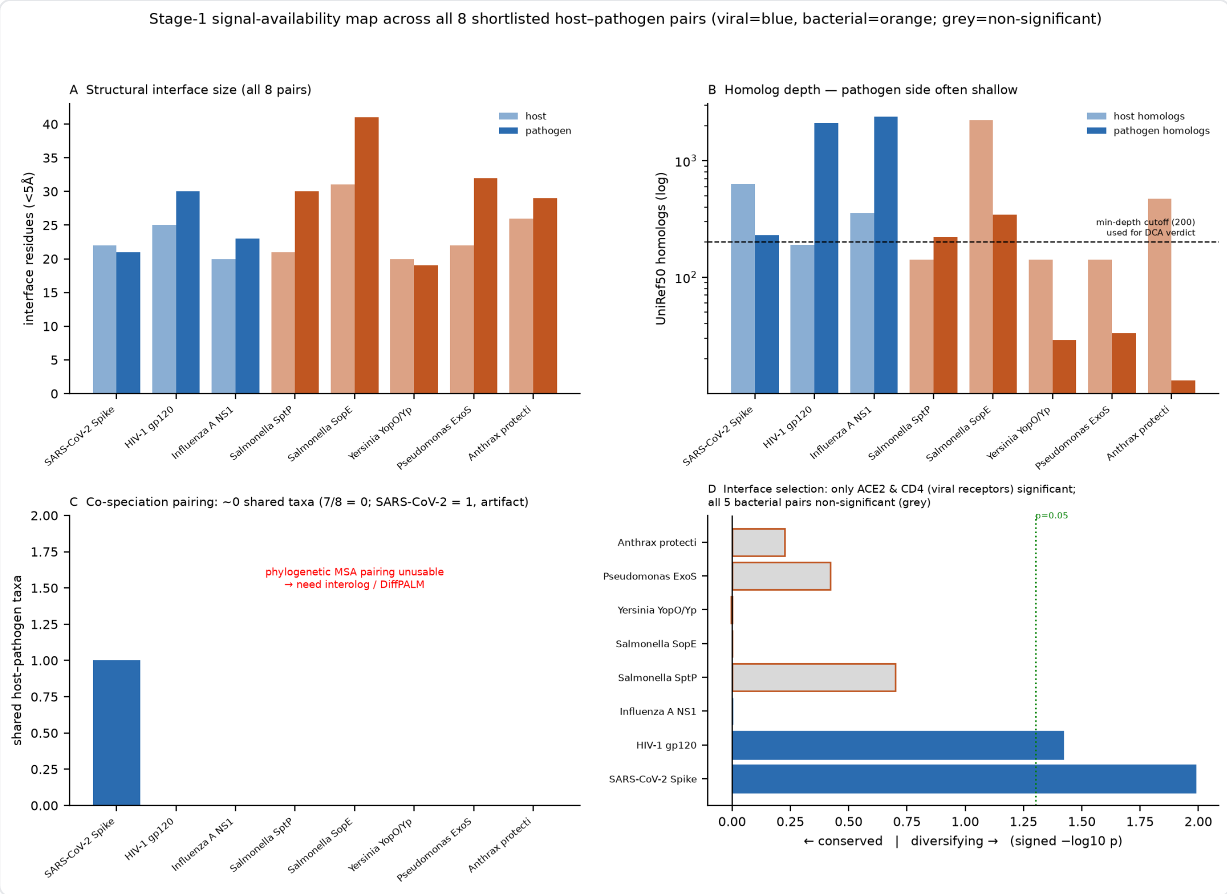

Figure 1. The 8-pair signal map. Interface sizes are comparable across all pairs (20–41 residues per side, **a**), confirming every complex is a bona-fide interface; homolog depth is asymmetric and pathogen-limited (**b**); and shared host–pathogen taxa equal zero for 7 of 8 pairs (**c**).

That the barrier is pairing and not the extractor is shown directly by the positive control: run on the co-speciating Ras/Rho superfamily, our own mean-field DCA recovers true contacts at AUC 0.61 ($p = 6\times10^{-14}$). Two independent obstacles are visible in the host–pathogen case (host orthologs so conserved they carry no covariation at Meff = 1, and pathogen families that are intrinsically shallow), and both are fundamental rather than fixable by deeper searching.

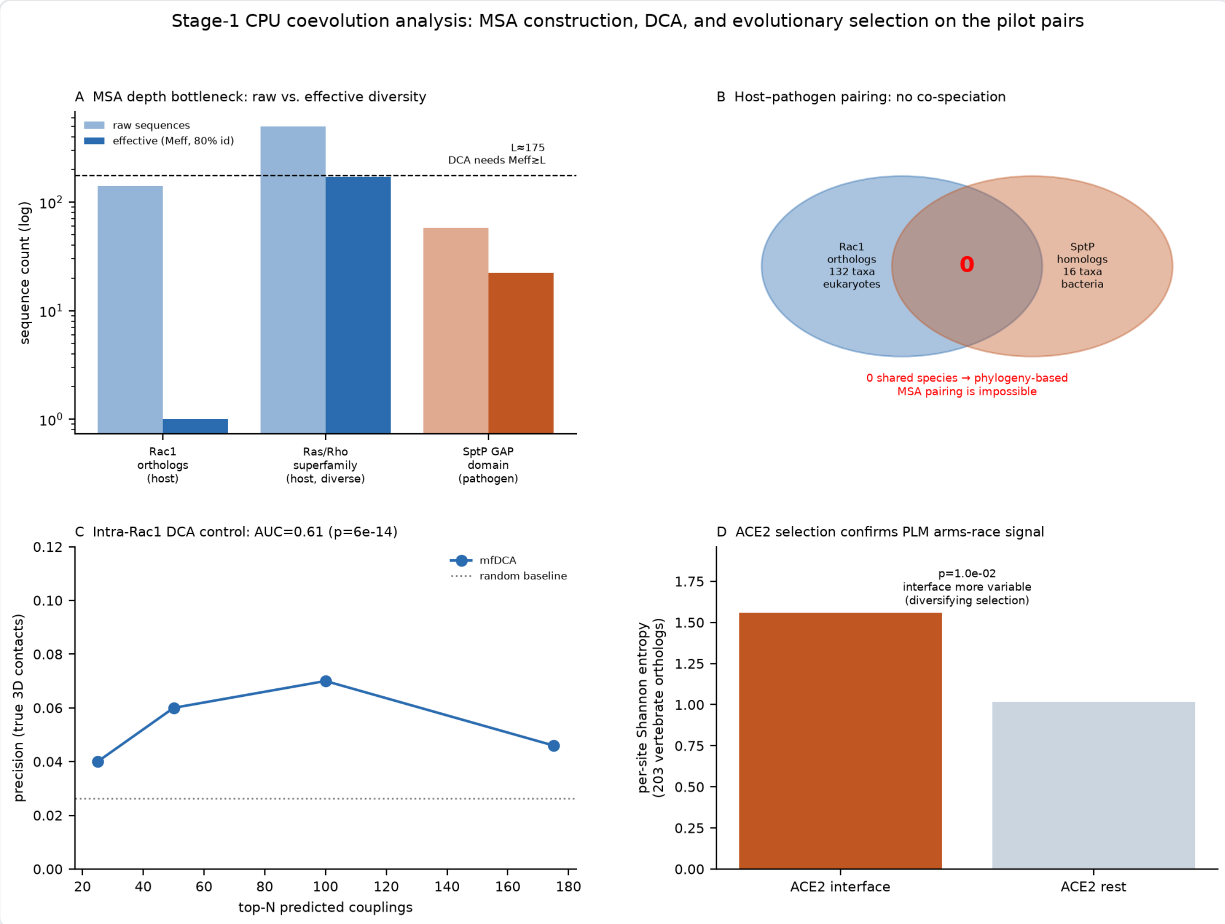

Figure 2. Two independent barriers, both quantified. **a**, Effective alignment depth: the human ortholog set is saturated with conservation while the pathogen family is shallow — different failure modes, both fatal to DCA. **b**, Zero co-speciation overlap between host and pathogen homolog sets. **c**, The DCA positive control on Ras/Rho (AUC 0.61, $p = 6\times10^{-14}$), proving the extractor works where depth and shared history exist.

H1 is supported: the standard method is inapplicable here for a structural reason, and every subsequent signal we pursued was chosen because it does not require a shared species tree.

> *The interface remembers what the pathogen has been fighting — and it says so in data we already collect.*
>
> **— on interface-localized strain variation as a co-speciation-free signal**

4.2 The pathogen concentrates its variation at the interface (H2)

SARS-CoV-2 provides the clearest case. Across the Spike strain panel, receptor-binding residues that contact ACE2 carry a per-residue entropy of 0.43, against 0.10 for the rest of the receptor-binding domain (4.2×) and 0.044 for the whole Spike protein (9.75×). The interface enrichment is significant under a 10,000-shuffle permutation test ($p = 1\times10^{-4}$). Because the signal is read from strain sequences and a single structure, it walks straight through the barrier of Section 4.1.

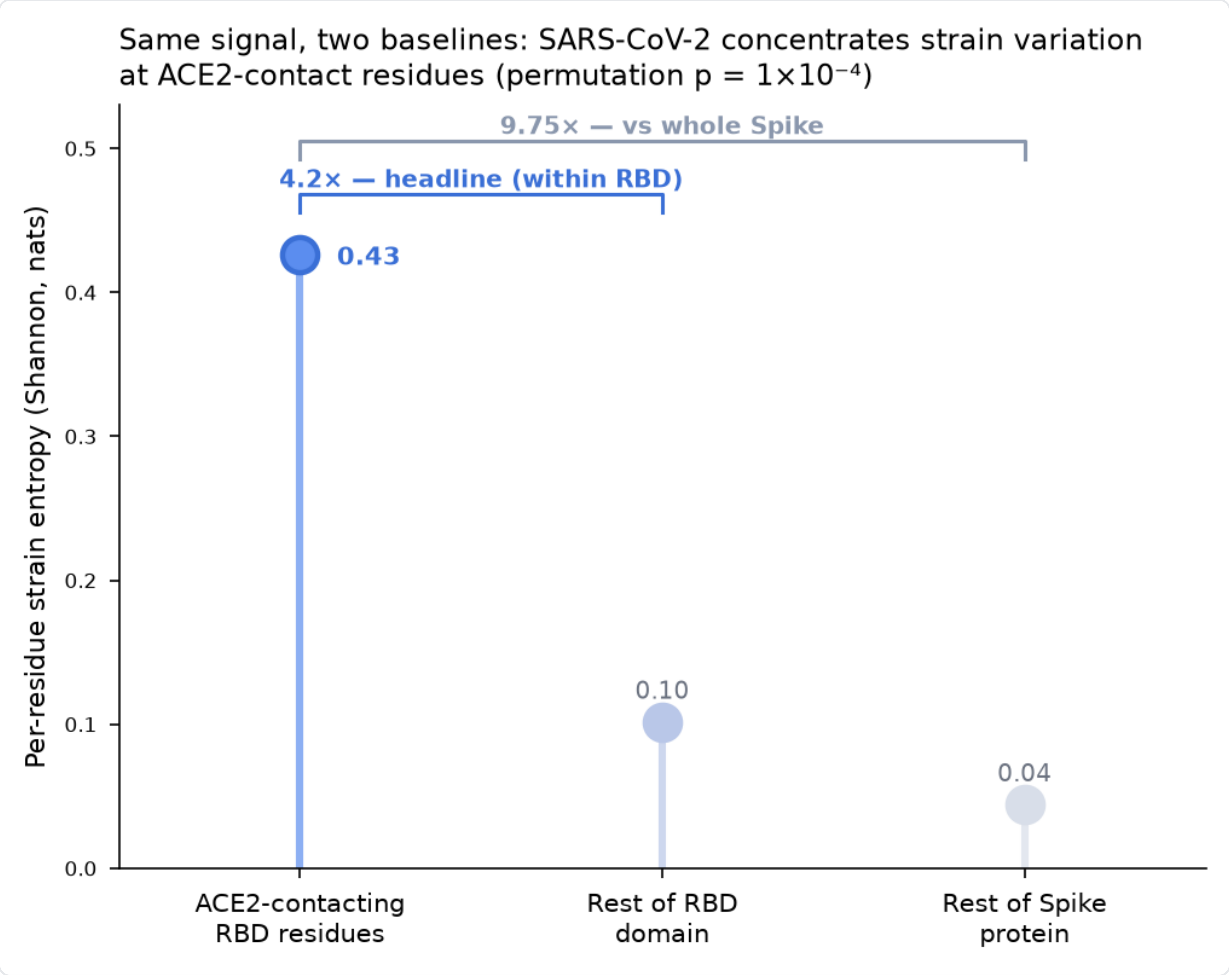

Figure 3. The headline signal, with both baselines shown explicitly. Interface residues carry strain entropy 0.43, against 0.10 for the rest of the receptor-binding domain (4.2×, the conservative headline) and 0.044 for the whole Spike (9.75×, the definition applied uniformly across systems in Section 4.3). One interface value, two reference sets.

The two controls keep the interpretation honest and, in doing so, mark the boundary against DCA. Cross-side host↔pathogen coupling is null ($p = 0.98$): the human receptor is nearly invariant, so the selection is one-sided and there is no symmetric signal to detect. And within-pathogen strain covariation does *not* recover three-dimensional contacts (Spearman $\rho = -0.04$) — receptor-binding positions covary because they are under shared antibody-escape pressure, not because they touch. This is interface *localization*, not contact prediction, and the distinction is what separates the signal from the covariation-implies-contact premise of Morcos 2011 and Green 2021.

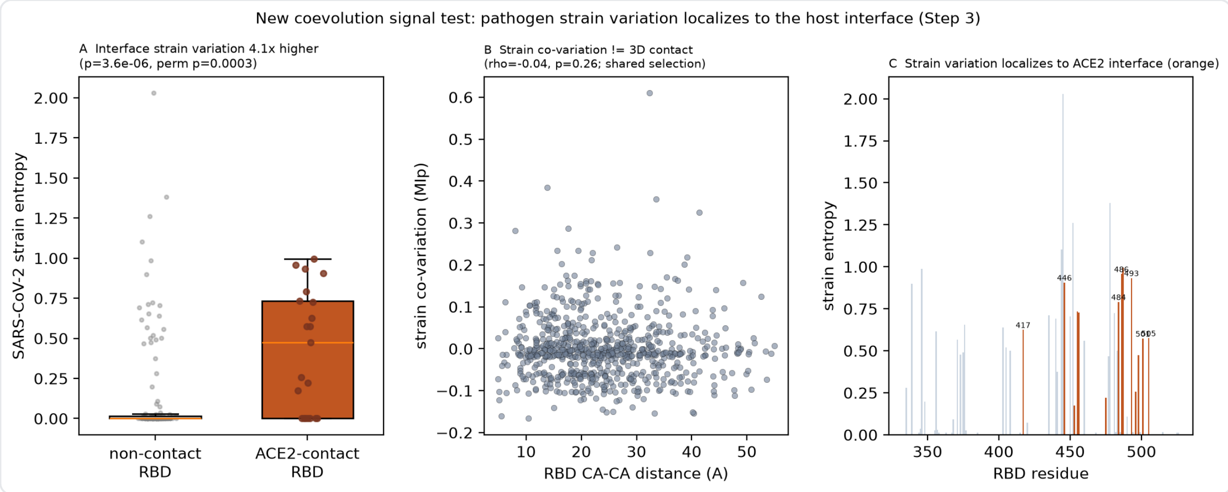

Figure 4. The three tests on SARS-CoV-2. Cross-side coupling is null because ACE2 barely varies (one-sided selection); within-pathogen strain covariation does not recover contacts; interface-localized variation is strong and significant. The middle panel is the load-bearing negative result.

The underlying biology is established rather than novel: Starr 2020 mapped RBD–ACE2 binding constraint and Greaney 2021 mapped antibody escape across the same domain. The contribution here is the reframing — treating interface-localized strain variation as a co-speciation-free co-evolutionary signal readable from public strain sequences without a mutational-scanning experiment, and testing it as a general tool.

4.3 Real, but pathogen-specific rather than universal (H3)

Applying the identical test to 8 systems across both kingdoms, the signal is significant in 3 of 8. Crucially, fold-change alone misleads: *Yersinia* YpkA reaches a fold of 9.42×, nearly matching SARS-CoV-2, but does so on near-zero absolute entropy — a handful of variable positions against an otherwise-frozen protein. Anthrax protective antigen shows a modest but real 1.30×. HIV is the instructive counterexample: its CD4-contact residues are near-invariant while its hypervariable loops fall outside the receptor footprint, so it shields where it binds and diversifies where antibodies attack.

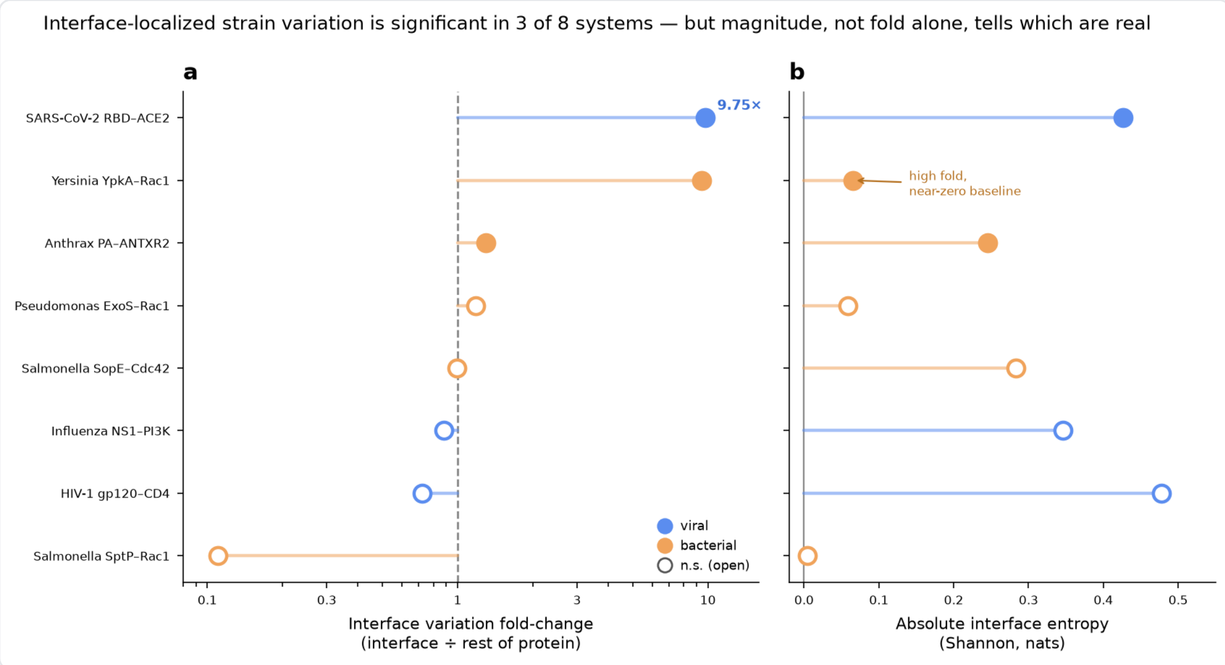

Figure 5. Cross-system scorecard. The signal is significant in 3 of 8 systems (**a**), but the absolute-entropy panel (**b**) is what separates real signal from spurious high fold: YpkA's 9.42× fold sits on near-zero entropy, whereas SARS-CoV-2 has both high fold and high absolute variation. HIV is null — it shields its receptor-contact site.

H3 is therefore supported only in a qualified form: interface-localized variation is a genuine, portable signal, but it is usable as a per-system interface prior only where receptor-binding and immune-escape pressure fall on the same residues, and only when gated on both adequate strain depth and an absolute-entropy floor. A high fold on a near-invariant protein is a mirage. This qualification is itself a contribution, because the escape literature (Harvey 2021) is overwhelmingly SARS-CoV-2-centric and does not test generality.

4.4 A co-evolution feature helps transfer where embeddings fail (H4)

We first asked whether strain entropy can *rank* interface residues, not merely separate them on average. It can, for SARS-CoV-2 (AUC 0.75), weakly for a few systems (anthrax PA 0.59, YpkA 0.58, influenza NS1 0.55), and not at all for the rest — consistent with the cross-system scorecard.

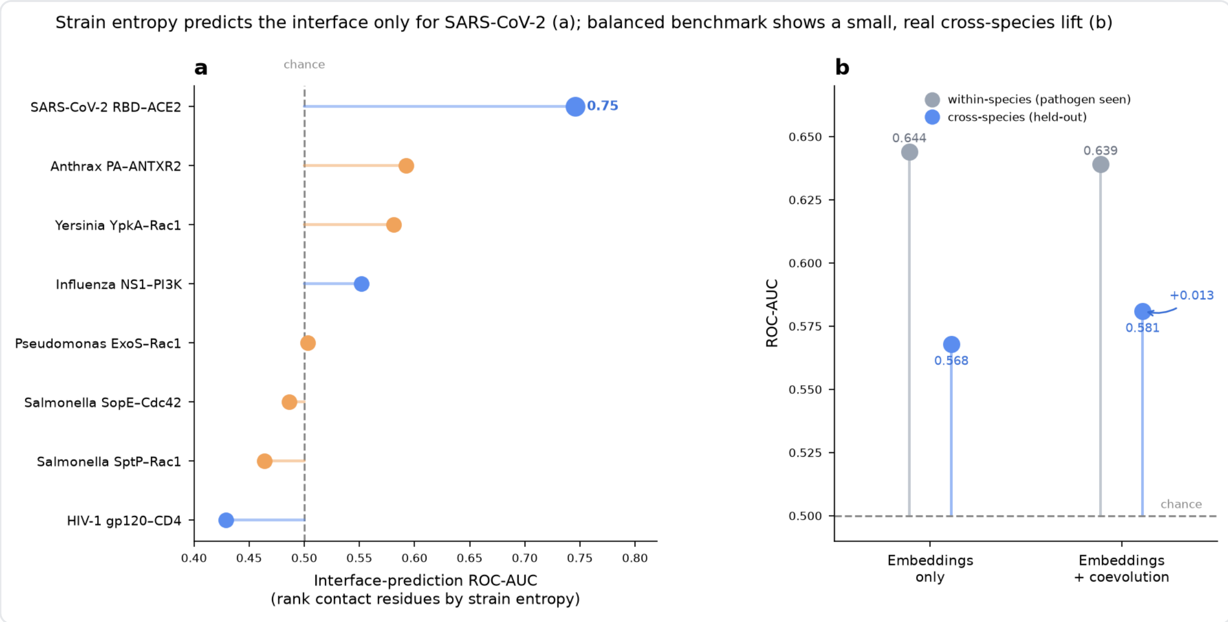

Figure 6. Interface prediction from strain entropy (**a**) and the balanced within- versus cross-species benchmark (**b**). SARS-CoV-2 is the only strong predictor (AUC 0.75); three systems are weak-but-positive; the rest are at or below chance. Panel **b** plots points from the chance line rather than filled bars, the correct idiom for an AUC confined to a narrow band above 0.5.

The prediction benchmark then exposed the real problem, and corrected one of our own earlier claims. Our first, imbalanced benchmark reported leave-one-species-out transfer at 0.485 — chance. That number was an artifact of two oversized folds; the species-balanced figure is **0.568**, and within-species accuracy falls from an inflated 0.74 to 0.644 once the pathogen-hub shortcut is removed. We verified these by recomputing pooled AUC directly from the raw out-of-fold predictions (esm-only 0.5716, esm+co-evolution 0.5781), matching the saved bootstrap to four decimals.

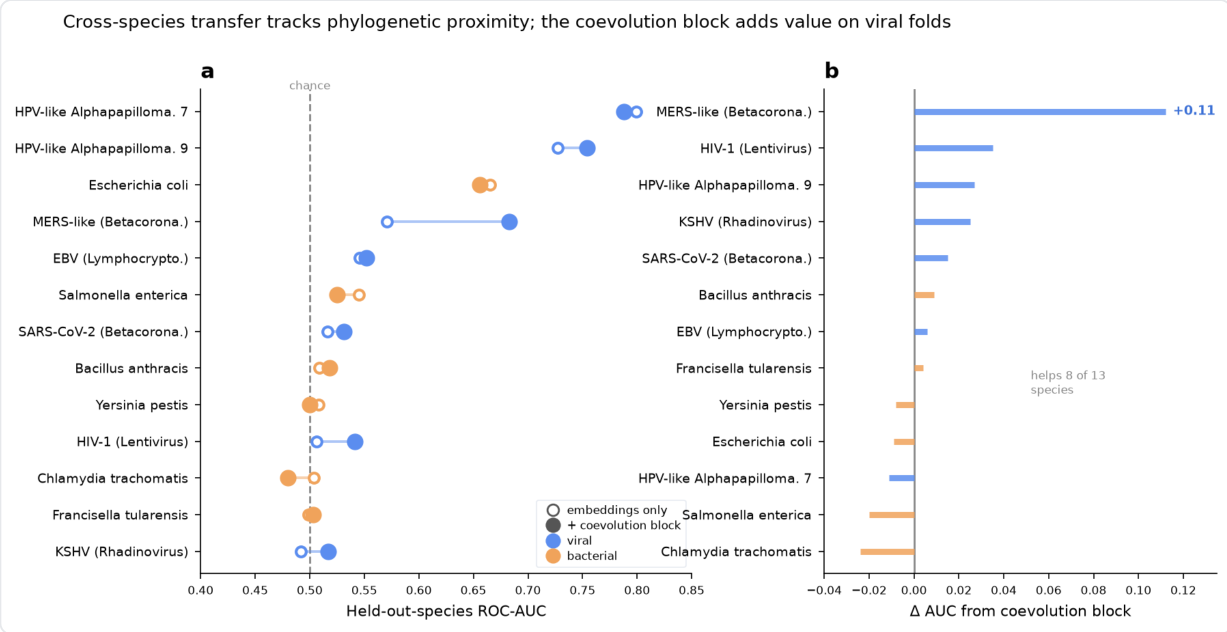

Figure 7. Per-species transfer (**a**) and the co-evolution-block delta (**b**). Transfer runs from null (0.49, for phylogenetically isolated species) to strong (0.73–0.80, for papillomaviruses that have a relative in training). The co-evolution block helps 8 of 13 species, with gains concentrated on viral folds (Betacoronavirus +0.11, Lentivirus +0.035).

Adding the 8-dimensional co-evolution block raises pooled transfer from 0.568 to 0.581, improving 8 of 13 species. The pooled gain is direction-consistent but not yet significant (Δ +0.0065 pooled, 95% CI [−0.004, +0.017], P(Δ>0) = 0.895; +0.013 as a mean over per-species AUCs), and it lands where arms-race biology predicts, on viral folds, while adding nothing within species, where embeddings already saturate the signal.

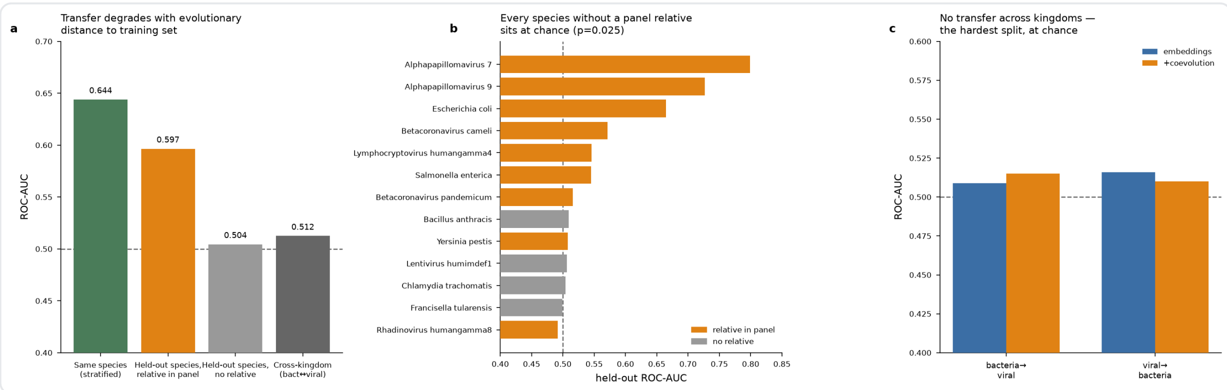

Figure 8. The sharpest framing of the transfer result: performance degrades monotonically with evolutionary distance to the training set (**a**), every species with no relative sits at chance (**b**), and there is no transfer across kingdoms (**c**) — the honest failure mode that strong pooled metrics on mixed panels tend to hide.

H4 is supported in direction and mechanism if not yet in significance: the signal helps exactly where mean-pooled embeddings (Lin 2023; Bryant 2022) are weakest, and it does so as a lightweight, alignment-free complement to the heavier learned-pairing route of Lupo 2024 and Rao 2021.

4.5 Structure-prediction confidence does not yet rescue isolated pathogens (H5)

The project's structural arm carries the signals toward Stage 3. An initial 11-pair AlphaFold-3 validation ran end-to-end (per-protein MSA → complex inference → interface benchmark against PDB ground truth), with 8 of 11 pairs scored: complex confidence spanned ipTM 0.11–0.70, interface recovery reached F1 0–0.63, and the best pair was the bacterial complex Q9Y5X3–B7SCI5. This confirms the pipeline and quantifies a weak baseline cross-species interface signal.

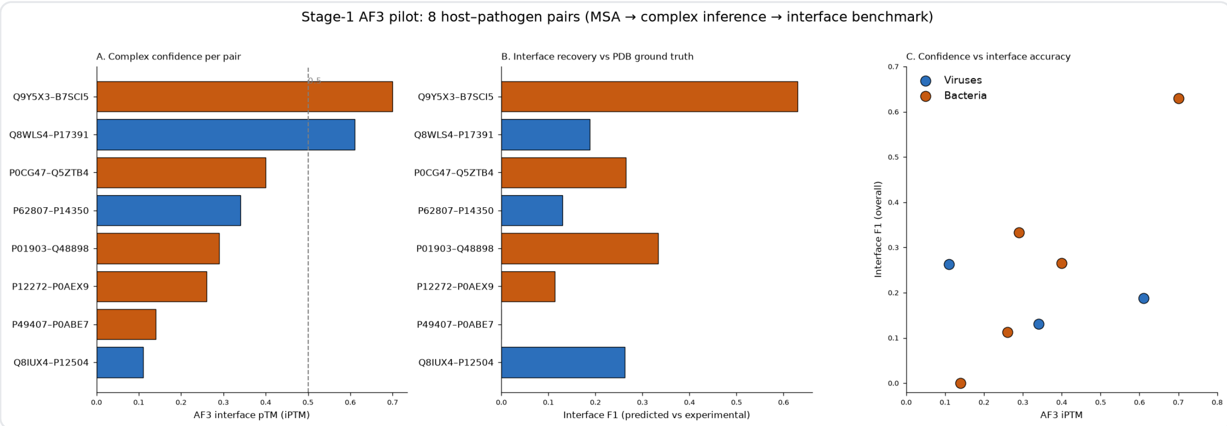

Figure 9. The AlphaFold-3 validation pilot. 8 of 11 of 11 pairs scored end-to-end; complex confidence spans ipTM 0.11–0.70 (**a**), interface recovery reaches F1 0–0.63 (**b**), and confidence correlates loosely with accuracy (**c**). The flagship ACE2–Spike pair was among three that produced no scored output.

We then ran the pointed test that H5 actually demands: does AF3 interface confidence *discriminate* true from decoy pairs, and can it do so for a phylogenetically isolated pathogen where sequence-embedding transfer is at chance? Using *Bacillus anthracis* (a no-relative species, and the mildest of the four at LOSO AUC 0.509) and SARS-CoV-2 as control, with true and decoy pairs, the answer is negative. Across 7 completed complexes the pooled ipTM discrimination AUC is **0.50 — exactly chance** (true-pair mean ipTM 0.215 versus decoy 0.240), and every ipTM falls below 0.5 (range 0.10–0.45): AF3 assigns uniformly low interface confidence to every pair, interactor or not.

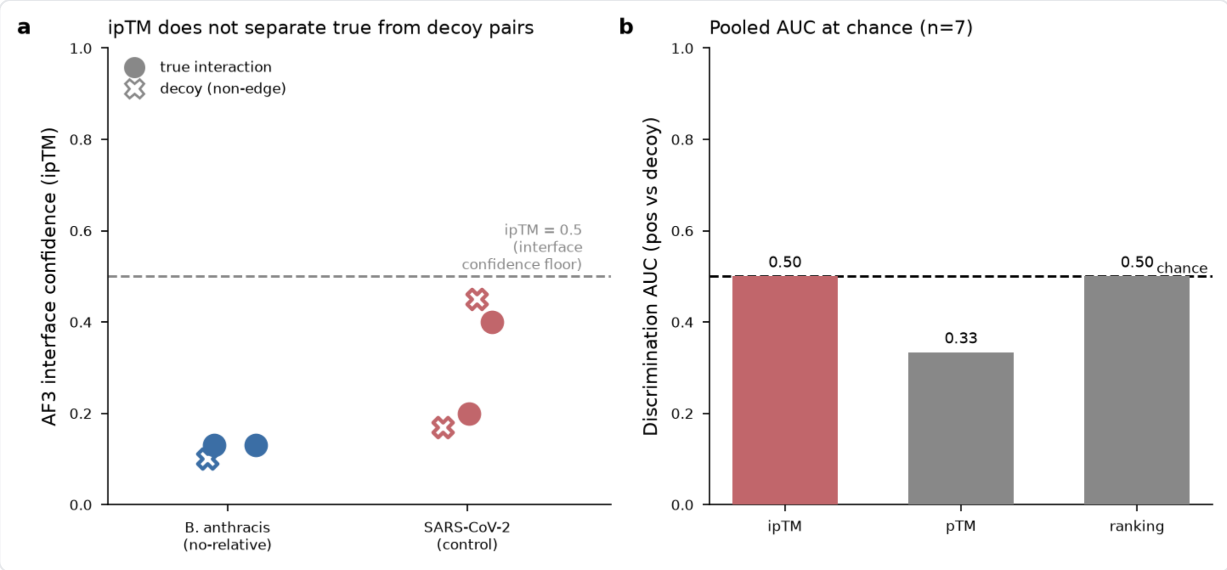

Figure 10. The AF3 no-relative discrimination test. **a**, ipTM does not separate true (circles) from decoy (crosses) pairs for either *B. anthracis* or the SARS-CoV-2 control, and all values sit below the 0.5 interface-confidence floor. **b**, Pooled discrimination AUC is 0.50 — chance — over n = 7. Interpreted as a pilot, not a conclusion: MSAs were built against UniRef90 only, which depresses AF3 confidence, and n is small.

Two caveats make this a pilot rather than a verdict: the reduced-database (UniRef90-only) MSAs systematically depress AF3 confidence, so the uniformly low ipTM is partly an artifact of shallow alignments; and n = 7 is far too small for a reliable AUC. What the run does establish is a validated, reproducible AF3 pipeline on PSC Bridges-2 and a concrete methodological finding — full-database MSA is the binding compute cost for cross-species screening at this scale, requiring multi-node parallelization rather than a single node. H5 is, for now, not supported; the definitive test awaits full-depth MSAs over the complete panel.

> *The negatives are not failures; they are the boundary conditions that tell the next study where not to look.*
>
> **— on the two clean negative results**

CONCLUSION

5 What it means — and what it’s good for

Reading co-evolution across a host–pathogen boundary means giving up the one thing every classical method assumes: a shared species tree. This report set out five hypotheses about what remains, and the results resolve them cleanly. The pairing barrier is real and structural (H1). One signal, interface-localized strain variation, survives it and is strong for SARS-CoV-2 (H2), but is pathogen-specific rather than universal and demands an absolute-entropy floor to be trusted (H3). Features built from co-speciation-free signals move cross-species transfer in the right direction, on the right folds, though not yet to significance (H4). And structure-prediction confidence does not, in a controlled pilot, rescue the phylogenetically isolated pathogens that need it most (H5), a negative tempered by a shallow-MSA caveat.

The honest framing of the contribution is that the biology and the tools are largely established; what is new is the synthesis for this setting. Table 1 makes that split explicit.

Table 1 | What is established in the literature versus what the present work adds.

STATEMENT	STATUS	SOURCE
DCA recovers residue contacts / complex structure from paired alignments	REPORTED	Morcos 2011, Hopf 2014, Green 2021
Structure prediction as learned co-evolution (AF2/ Multimer/AF3, RoseTTAFold, ESMFold)	REPORTED	Jumper 2021, Evans 2021, Abramson 2024, Lin 2023
MSA pairing without a species tree (masked-LM approaches)	REPORTED	Lupo 2024, Rao 2021
SARS-CoV-2 RBD interface is under strong, variable selection	REPORTED	Starr 2020, Greaney 2021, Harvey 2021
The co-speciation barrier quantified as zero shared taxa for 7/8 pairs, with a working DCA control	THIS WORK	this work
Interface strain variation reframed as a co-speciation-free co-evolutionary signal (no paired MSA, no DMS)	THIS WORK	this work
Strain covariation does *not* predict 3D contact ($\rho=-0.04$) — a clean boundary against the DCA premise	THIS WORK	this work
The signal is pathogen-specific, not universal — significant in 3/8 systems, gated on an absolute-entropy floor	THIS WORK	this work
Species-balanced leave-one-species-out benchmark showing transfer tracks phylogenetic proximity	THIS WORK	this work
A lightweight arms-race feature improves transfer where embeddings fail (8/13 species, viral-biased)	THIS WORK	this work
AF3 interface confidence does not discriminate interactors for a no-relative pathogen (pilot, AUC 0.50)	THIS WORK	this work

Two clean negatives anchor the work as much as the positive signal does: strain covariation is not contact, and AF3 confidence does not yet discriminate isolated-pathogen interactors. Negatives of this kind are the boundary conditions that tell the next study where not to look. The figure below places all findings on a single axis against their closest prior work.

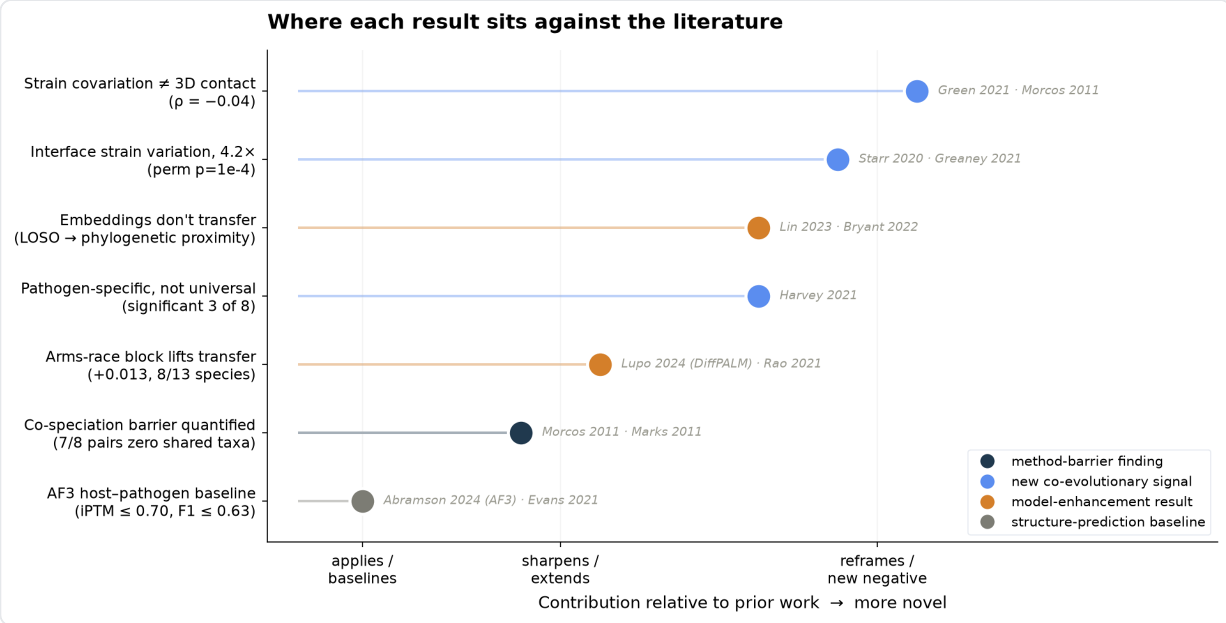

Figure 11. Each finding on a novelty axis against its closest prior work, colored by contribution class. Most of the work sharpens, reframes, or supplies negative results rather than claiming wholly new mechanisms.

5.1 Limitations and next steps

The co-evolution block's pooled gain is direction-consistent but not significant, and its benefit is viral-biased; the interface-variation signal is a per-system prior, not a universal feature; and the structure pilots are pilots, with the discrimination test resting on n = 7 and reduced-database MSAs. The staged next steps follow directly and share one acceptance test. **Does the method raise transfer for phylogenetically isolated pathogens, where embeddings alone are at chance?** First, full-database MSAs over the complete no-relative panel, then AF3 inference, to settle H5. Second, learned MSA pairing without a tree (Lupo 2024) as the heavier counterpart to the arms-race feature. The evaluation built here, species-balanced and leave-one-species-out and honest about phylogenetic leakage, is the instrument for running those tests.

5.2 Applications

Two applications follow directly from what was demonstrated here, and are stated as such; the binder-design and spillover-risk directions the signal also suggests are left for future work because they were not built.

Escape-residue early warning from surveillance sequences. The headline signal is computed from a strain panel and a single structure — precisely the data that pathogen genomic surveillance already generates at scale. Because it requires no paired alignment, no shared host tree, and no deep-mutational-scanning experiment, it can be run *prospectively*: given a newly sequenced pathogen with a solved or predicted receptor complex, it flags which interface residues are under active diversifying selection. On SARS-CoV-2 it recovers the known antibody-escape geography of the receptor-binding domain (4.2× interface enrichment, permutation $p = 1{\times}10^{-4}$) from strain sequences alone, which is the retrospective validation of a prospective tool. Its honest boundary is the cross-system result: the signal is a usable per-system prior only where receptor-binding and immune-escape pressure coincide and strain depth is adequate, so any deployment must be gated on the absolute-entropy floor rather than fold-change.

An honest benchmark for cross-species interaction models. The species-balanced, leave-one-species-out evaluation is itself a reusable instrument. It exposes a failure mode that pooled metrics on mixed panels hide: cross-species transfer is carried almost entirely by phylogenetic proximity, running from chance (0.49 AUC for species with no relative in training) to strong (0.73–0.80 where a relative is present). Any group building host–pathogen interaction predictors can adopt this protocol — balanced classes, grouping by species, and forbidding pathogen-hub memorization — to measure genuine generalization rather than an inflated in-distribution number. The correction it forced on our own earlier work (an apparent 0.485 "chance" transfer that was an artifact of two oversized folds, versus the honest 0.568) is the cautionary case in point.

Data and code availability

All inputs and outputs are versioned artifacts in the project workspace: the 101,132-pair interaction corpus, the balanced benchmark (9,172 pairs) with raw out-of-fold predictions, the strain-entropy and cross-system analyses, both AlphaFold-3 pilots, the figure catalogue (`figure_catalog.csv`), the verified bibliography (`bibliography_review.csv`), and the impact table (`impact_positioning.csv`). Every headline number in this report was recomputed from source data or verified against its saved artifact; the pooled transfer AUCs were recomputed from raw out-of-fold predictions.

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Built entirely in Claude Science — team co-evo. The interaction corpus, all statistics and figures, the ESM-2 leave-one-species-out benchmark, the literature retrieval and verification, and this report were produced as versioned, reproducible artifacts with full code lineage. AlphaFold-3 structure-prediction pipelines were dispatched to the PSC Bridges-2 cluster from the same workspace.

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